

VARUN K S

AI Engineer • LLM & RAG Application Developer

+91 9600785828 | itsvarun310@gmail.com | Coimbatore, Tamil Nadu

PROFESSIONAL SUMMARY

AI Engineer specializing in LLM application development, Retrieval-Augmented Generation (RAG), and end-to-end conversational AI systems, with additional depth in computer vision. Experienced in building and deploying LLM-driven products — from natural-language data platforms and domain-specific chatbots to autonomous agentic systems — using tools such as LangChain, Gemini API, and n8n. Proven track record at national and international hackathons, including SIH 2024 Winner and Top 2% Finalist at the India AI Impact Buildathon. Passionate about translating cutting-edge LLM research into practical, production-ready AI systems.

TECHNICAL SKILLS

LLMs & AI Engineering	LLM Application Development • RAG Pipelines • LangChain • Prompt Engineering • Agentic Systems • LLM Inference (Local & API)
AI / ML	Machine Learning • Deep Learning • NLP • Computer Vision
Frameworks	TensorFlow / Keras • scikit-learn • XGBoost • OpenCV • PyTorch
Languages	Python • SQL • C
Tools & Automation	n8n (Workflow Automation) • Git
Soft Skills	Team Leadership • Communication • Research & Documentation

EDUCATION

B.Tech – Artificial Intelligence & Machine Learning | Bannari Amman Institute of Technology, Coimbatore
Mar 2026

PROJECTS

Dhurvam – AI-Powered Honeypot System

- Engineered an AI-powered honeypot that detects scam messages, engages scammers in convincing believable conversations, and extracts actionable intelligence — built for the GUVI AI Hackathon (India AI Impact Buildathon 2026 — Top 2% National Finalist).
- Leveraged LLM-based dialogue management to simulate realistic victim behavior, enabling intelligence gathering on scam tactics and actors.

GLOF Early Warning System – SAR Image Analysis

- Designed a predictive risk-detection system using Sentinel-1 SAR satellite imagery and CNN models to identify early-stage Glacial Lake Outburst Flood (GLOF) indicators.
- Achieved high spatial accuracy by fine-tuning convolutional architectures on geospatial remote sensing data.

GLOF Early Warning System – IoT Sensor Network *(SIH 2024 Winner)*

- Developed a real-time GLOF monitoring system integrating water level, temperature, ground motion, pressure, and flow rate IoT sensors with XGBoost for anomaly detection.
- Won Smart India Hackathon 2024 (problem statement by DRDO); system designed for deployment in Himalayan glacial zones.

Samudra (FloatChat) – Conversational Ocean Data Platform

- Built an AI-powered conversational platform that transforms access to ARGO ocean data by converting complex NetCDF files into user-friendly, searchable, and visual formats through LLM-driven natural language queries.
- Automated data ingestion, advanced analytics, and interactive dashboard visualizations, making oceanographic insights accessible for both experts and non-specialists.

Solace – AI Mental Health Chatbot

- Built an emotionally intelligent mental health chatbot using Google Gemini with real-time NLP, sentiment analysis, and dynamic user personalization for empathetic, context-aware responses.

Scream – Real-Time Scream Detection

- Developed a real-time audio classification application using MFCC feature extraction combined with SVM and MLPClassifier models, achieving high accuracy in distinguishing screams from ambient noise.

Aran – Local Vulnerability Report Generator

- Built a local LLM-inference-powered report generator for vulnerability analysis — Finalist at the Israel-India Global Innovators Hackathon.

Neethi – AI Legal Assistance Chatbot *(Final Year Project)*

- Designed an AI-powered conversational assistant to help Indian citizens navigate legal services and access information about various Department of Justice initiatives.
- Leveraged RAG (Retrieval-Augmented Generation) technology with local LLM integration to deliver accurate, contextual responses.

ACHIEVEMENTS & EVENTS

- Smart India Hackathon 2024 — Winner: Led ML/DL development for GLOF early warning system; problem statement provided by DRDO.
- India AI Impact Buildathon by HCL GUVI — Top 2% National Finalist: Part of AI Impact Summit 2026; built Dhurvam, an AI honeypot system for scam detection and intelligence extraction.
- Israel-India Global Innovators Hackathon — Finalist: Developed Aran, a local LLM-based vulnerability analysis report generator.
- 4th International Conference on Engineering & Advancement in Technology: Presented a research paper on 'Advancement in Nanostructured Materials for Sustainable Energy Harvesting and Storage'.

CERTIFICATIONS

- Google Cloud Skills Boost: Introduction to Generative AI
- Google Cloud Skills Boost: Introduction to Large Language Models
- Google Cloud Skills Boost: Transformer Models and BERT Model
- Coursera (UST): Create Your First Python Program
- Cognitive Class: Python 101 for Data Science